MARTIN HUBER, PHD

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SUMMARY

Robotics software machine learning engineer (C++, Python, CUDA) with 5+ years of experience in hard real-time & multi-threaded systems, computer vision, multimodal data processing, and autoregressive imitation learning. Proven track record of delivering robust, production-grade autonomy software stacks across various robotic systems, alongside technical leadership and mentoring.

EXPERIENCE

Research Assistant / Associate in Surgical Robotics Oct 2019 - Mar 2022 / Nov 2022 - Mar 2026
King's College London *London, UK*

- Developed and maintained open-source C++ ROS 2 control robotics software stack incorporating force control (260+ stars on GitHub: link).
- Developed a differentiable rendering-based localization of surgical robots via transformer-based segmentation models (link to deployment video, 100+ stars on GitHub: link).
- Published in peer-reviewed journals and conferences in robotics and computer vision on autoregressive imitation learning for autonomous robotic surgery.

Robotics Software Engineer Apr 2022 - Oct 2022
Johnson & Johnson, MedTech *Redwood City, CA, USA*

- Implemented 1000x faster multithreaded manipulability search in production for Monarch robot.
- Developed CUDA-accelerated forward kinematics kernels with abstracted device memory handling.
- Enhanced core algorithms (forward and inverse kinematics) and added extensive test coverage to production.
- Participated in code reviews, bug fixes, built Isaac Sim URDF, and contributed to ROS 2 integration.

Research Assistant Aug 2016 – Mar 2017
German Cancer Research Center *Heidelberg, Germany*

- Developed a generative AI-based inpainting method for CT scans, enabling 75% acquisition dose reduction while preserving diagnostic image quality (link to GitHub).

SKILLS

Technical Skills	Software development, CI/CD, robotics, multimodal data, computer vision, imitation learning, system integration
Platforms & Systems	NVIDIA Jetson edge & hard real-time, distributed deep learning, Linux, ROS 2
Programming & Tools	C++, Python, CUDA, CMake, PyTorch, Git, GitHub, Docker, OpenCV

FIRST AUTHOR PUBLICATIONS

Huber, Martin et al. LBR-Stack: ROS 2 and Python integration of KUKA FRI for Med and IIWA robots. *Journal of Open Source Software*, 2024.

Huber, Martin et al. Localising under the drape: Proprioception in the era of distributed surgical robotic systems. *arXiv*, 2025.

Huber, Martin et al. Hydra: Marker-free RGB-D hand–eye calibration. *arXiv*, 2025.

Huber, Martin et al. Deep homography estimation in dynamic surgical scenes for laparoscopic camera motion extraction. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, 2022.

Huber, Martin et al. Deep homography prediction for endoscopic camera motion imitation learning. *MICCAI*, 2023.

Huber, Martin et al. Homography-based visual servoing with remote center of motion for semi-autonomous robotic endoscope manipulation. *ISMR*, 2021.

TECHNICAL LEADERSHIP & MENTORING

PhD Biomedical Engineering (2025) Guided streamlining of stereo differentiable rendering via CUDA streams for marker-free online tracking of industrial manipulators with *publication in IJCARS*

MSc Healthcare Engineering (2025) Guided development and deployment of kinematic remote center of motion.

Engineering Intern (2025) Guided ROS 2 control integration of Dynamixel motors via microcontroller in C++.

Engineering Intern (2025) Mentored development of a robotic tool adaptor with *publication at HSMR*.

MSc Computer Science (2019) Mentored on-device Google MediaPipe hand tracking for robot control in Gazebo.

BSc Physics (2018) Guided WaveNet autoregression of direct multiple shooting solutions for differential equations.

EDUCATION

PhD in Biomedical Engineering & Imaging Sciences, King's College London 2019 - 2024
Thesis: Data-Driven Robotic Endoscope Automation. Fully funded under European Union's H2020 grant agreement No 101016985.

MSc in Physics, Heidelberg University 2016 - 2019
Relevant Coursework: Machine learning for computer vision (4.0 / 4.0 GPA - First-Class equivalent), artificial intelligence (4.0 / 4.0 GPA - First-Class equivalent). Thesis: Non-linear model predictive control for humanoid gait generation and autonomous control via imitation learning (link to GitHub).

AWARDS & GRANTS

NVIDIA Academic Grant. Awarded GPUs worth £21,000. 2x NVIDIA RTX PRO 6000 Blackwell Max-Q Workstation Edition and 1x Jetson AGX Orin for developing Spatially Intelligent Surgical Robots.

Best Poster Award. Open-Source Software for Surgical Technologies at the Hamlyn Symposium 2024.

PROJECTS

Open Source Contributions. Downstream maintainer of the hard real-time ROS 2 control hardware abstraction layer. Built a ROS 2 Large Language Model (LLM) interface (link to GitHub).

PPO Reinforcement Learning in C++. Implemented Proximal Policy Optimization with Libtorch for efficient deployment on real-time robotic systems (link to GitHub).

TEACHING

Scientific Programming, MSc Healthcare Technology Sep 2024 – Dec 2024
King's College London *London, UK*

- Delivered lectures on introduction to Python for scientific programming.

Robotics I, BSc Computer Science Nov 2018 – Mar 2019
Heidelberg University *Heidelberg, Germany*

- Authored a comprehensive tutorial script covering core robotics concepts (link to manuscript).
- Delivered weekly tutorials and developed accompanying exercises for undergraduate computer science students.